

Acronym:

TCS = Thermal Control System

1.5.5 Thermal Management Subsystem

1.5.5.1 Thermal Management Subsystem Requirements

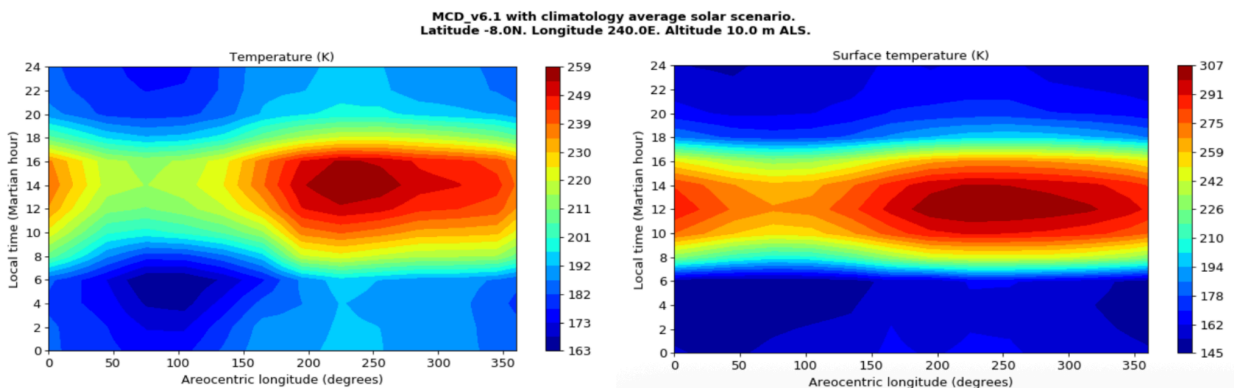
Req #	Requirement	Rationale	Parent Req.	Child Req.	Verification Method	Req Met?
TCS-01	The TCS shall maintain a consistent internal system temperature within component operational temperatures while exposed to ground temperatures within the range 128K - 324K and air temperatures in the range 149K - 272K	The system must remain within operational temperatures at all times, and this range was chosen based on data from the Mars Climate Database with a 10% added buffer to account for variance and other factors.	SYS-01	TCS-03 TCS-04	Test	Met, shown by heat flow maps
TCS-02	All purchased components, manufacturing, testing facilities, and other costs shall total less than TBD dollars	Overall cost must remain within budget, and TBD dollars have been allocated to the completion of the TCS.	MR-09		Inspection	Met
TCS-03	The TCS shall be capable of sensing internal temperature	Readings from temperature sensors inform the TCS how much heat to add to the system to maintain operational ranges for all subsystems	TCS-01	TCS-05	Inspection	Met
TCS-04	All heat generation hardware shall have at least one redundant component	Any component overheating or freezing would inhibit it from performing its intended function, so the consequences of a heating component failing are significant.	TCS - 01		Inspection	Met
TCS-05	At least 3 temperature sensors shall be present in the system	Adds redundancy while showing where heat may need to be directed	TCS-03		Inspection	Met

TCS-06	Power consumption by the TCS shall not exceed TBD Watts	Power is needed for all operations, so maintaining TCS power consumption below a specified threshold allows other systems to utilize consistent power levels for the completion of their tasks	SYS-05	TCS-07	Inspection	Met
TCS-07	Passive heating techniques shall be utilized within the TCS	Passive heating techniques (such as insulation and coatings) do not take power, so using them will minimize overall power usage	TCS-06		Inspection	Met
TCS-08	The TCS shall minimize its mass and volume	The overall system has strict mass and stowed volume constraints, so minimizing TCS mass and volume can allow for more to be used elsewhere	MR - 05		Inspection	Met

TCS-01 defines the overall goal of the system, ensuring that it can accomplish its primary task. The air and ground temperature ranges chosen are based on the following data (Mars Climate Database).

Reasoning for TCS-01 range

- Ability to maintain consistent internal temperature for a ground thermal range of 128K to 324K and an air temperature range of 149K to 272K



- Data From [Mars Climate Database](#)

- A buffer of 10% is added to the ranges seen in this data to increase the factor of safety to account for variations in location and local temperature and other unexpected variations
- If the system operates effectively at both extremes of temperature, we can assume that it would also operate effectively at temperatures within the range

1.5.5.2 Thermal Management Subsystem Overview.

The thermal control subsystem ensures that all the systems remain within the operational temperature ranges by controlling heat inputs, heat transport, storage, and ultimately rejection. It utilizes passive and active instruments to maintain the temperature. The TCS will primarily consist of three potential subassemblies: heat generation, heat distribution, and environmental interface. Heat generation creates the heat needed to maintain component temperatures during cold periods through electrical and chemical means. The heat distribution subsystem transfers heat from high-power locations to a place where it can be stored or rejected efficiently. The environmental interface controls the radiative and conductive exchange with the external environment through tools such as insulation and thermal coatings.

Heat Generation Subsystem

Function and Objectives

The heat-generation subsystem provides active temperature control during low-temperature exposure, where dissipation internally is insufficient to maintain operating temperatures. Its purpose is to prevent batteries, propellant valves, sensors, and electronics from dropping below their lowest survival levels. Because Martian surface exposure to the sun's rays is low and nights are long and cold, the subsystem must be operational to maintain all instruments in an operational-ready state.

Sub-Components

- Patch heaters are laminated aluminum or composite surfaces with polyimide-film resistance elements. When energized, they generate heat linearly to the applied voltage and can be modulated continuously. A single patch ($\sim 50 \times 50$ mm, 1 mm thick) delivers up to 10 W at 28 VDC.
- Thermostats and Control Relays: Two-channel thermostats use a bimetallic that interfaces with solid-state relays. Fail-safe configurations ensure that a single failure will not prevent heater operation when required.

- Solid-state controllers: Utilize temperature sensors to read the temperatures and send signals to relay controlled heaters to turn on or off as required.

Design Performance

The thermal system maintains internal structural temperatures between 250 K and 310 K even under uniform external ambient conditions as low as 128 K. Redundancy solves the requirement under TCS-04, ensuring no single heater failure jeopardizes subsystem operability. Total heater power usage is restricted below TBD, consistent with TCS-06.

Technology Readiness Level

All heating elements and control units employ tried and tested designs qualified for space use.

- Patch Heaters: TRL 7
- Thermostats and relays: TRL 7
- Solid-state controllers: TRL 7

Overall subassembly TRL:7

Heat Distribution Subsystem

Function and Objectives

The heat-distribution subsystem regulates the distribution of thermal energy throughout the spacecraft, removing waste heat from power electronics, supplying it to establish uniform internal temperatures, and supplying excess heat to the environmental interface for removal.

Sub-Components

- Thermal Straps: Flexible, high-conductivity connections transfer heat away from electronics, structural panels, and radiator bases. Avionics and structural interfaces use braided aluminum straps. Low-mass, high-conductivity path pyrolytic graphite sheet (PGS) straps are used near sensitive payloads. They end with bolted or clamped aluminum end-fittings to provide isothermal contact.

Design Performance

It has zero power consumption, no startup sequence or warm-up circuits, low mass and volume, no fluids or moving parts, and flexible mounting.

Technology Readiness Level

This technology has been used in prior space missions and has been in use for the last few decades.

Thermal Straps: TRL 6

Overall subassembly TRL: 6

Environmental Interface Subsystem

Function and Objectives

The environmental-interface subsystem is responsible for regulating heat exchange between the environment and the spacecraft. Both active radiative surfaces and passive insulation are used to regulate the inflow and outflow of heat. This subsystem is essential for maintaining the internal energy composition of the spacecraft as well as meeting requirement TCS-07 (passive heating techniques).

Sub-Components

- A white-painted radiator on the top face, to reject heat to deep space
- Multi-Layer Insulation (MLI) over all other surfaces, for environmental protection and heat retention.

Design Performance

Analytical models indicate that the Multi-Layer Insulation reduces exterior heat gain and the radiators dissipate all steady-state waste heat without going over structural temperature limits.

Technology Readiness Level

White radiator: TRL 8

MLI: TRL 7

Overall subassembly TRL: 7

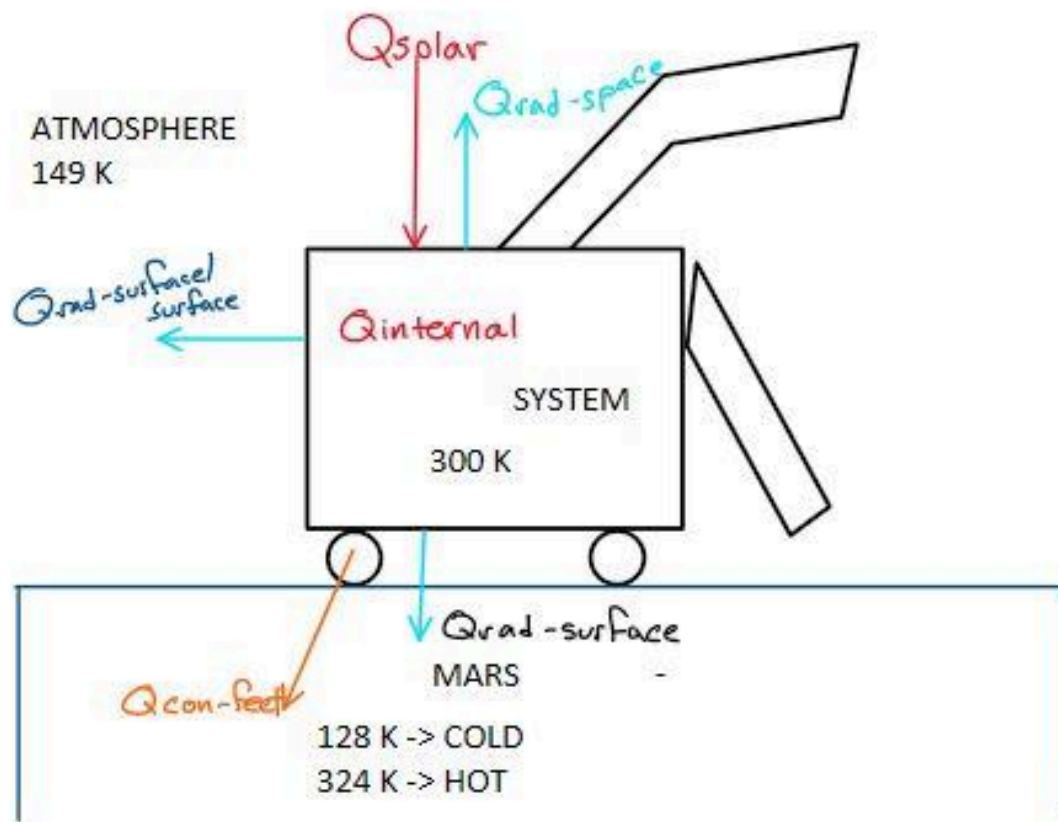
System Integration and Interfaces

The three subassemblies are mechanically and thermally connected through a support structure and harnesses.

- **Heat Flow Integration:**
Heat generated by electronic components is gathered by the distribution network, transported to the radiator panels, and rejected to space through the environmental interface.
- **Power Integration:**
Heaters draw power from the main Power Distribution Unit (PDU) under software control to avoid the total TCS power from exceeding the allocated budget (TCS-06).
- **Data Integration:**
The solid-state controller communicates with the CDH bus for command logic and temperature telemetry.

Across the thermal subsystem, the TRL of components and subassemblies is quite high. These high TRL designations are due to the proven uses of similar components and subassemblies in previous missions. They were not given the maximum TRL designation of 9 because they will be used in somewhat different applications than in past missions. Overall, the thermal control system for this mission will use many heritage components which were successful in previous missions, but additional testing and simulation is necessary to raise the TRL of these components and subassemblies for this mission to an acceptable level before launch.

Overall System TRL: 6



Assume:

- Major surface area sizes 0.5 m^2
- 110 W internal heat load
- Aluminum 6061
- White paint covers top surface and MLI covers all external surfaces

HOT CASE

$Q_{\text{solar}} = 56.05 \text{ W}$ (painted white)
 $Q_{\text{internal}} = 110 \text{ W}$
 $Q_{\text{rad-space}} = 172.53 \text{ W}$ (painted white)
 $Q_{\text{rad-space/surface}} = 1.33 \text{ W}$ (Wrapped in MLI)
 $Q_{\text{rad-surface}} = -0.414 \text{ W}$

$Q_{\text{in}} = 166.46 \text{ W}$
 $Q_{\text{out}} = 167.21 \text{ W}$
 $Q_{\text{net}} = -0.75 \text{ W}$

COLD CASE

$Q_{\text{solar}} = 0 \text{ W}$ (painted white)
 $Q_{\text{internal}} = 110 \text{ W}$
 $Q_{\text{rad-space}} = 172.53 \text{ W}$ (painted white)
 $Q_{\text{rad-space/surface}} = 2.16 \text{ W}$ (Wrapped in MLI)
 $Q_{\text{rad-surface}} = 1.110 \text{ W}$

$Q_{\text{in}} = 110 \text{ W}$
 $Q_{\text{out}} = 182.27 \text{ W}$
 $Q_{\text{net}} = -72.27 \text{ W}$

Subassembly	Mass	Dimension	Power Draw
Heat Generation	1 kg	Patch heater: 10mm x 10mm Solid-state controllers: (89 × 63 × 54 mm)	75 W (max)
Heat Distribution	10 kg	~3 meters long	N/A
Environmental Interface	Negligible	N/A	N/A

1.5.5.3 Thermal Management Subsystem Trade Studies (2-4 pages recommended)

Provide the trade studies your team performed when making design decisions for all subassemblies of the thermal subsystem. A narrative should accompany each trade study to describe it. The narrative should include why the weights were chosen, why the scoring was assigned, and the end result.

[Trade Studies Google Sheet](#) (images below)

Prior to performing trade studies, the TCS was split into three potential subassemblies: heat generation, heat distribution, and environmental interface. Heat generation is the only of the three that is known to be necessary, so the other trade studies both included one option which is to not design anything specifically for that subassembly.

Heat Generation Method						
Criteria	Explanation	Grade	Weight: Scale {1-5}	Patch Heaters	Cartridge Heaters	Deployable Surface to Increase Solar Flux
Cost	TCS - 02	10 = cheap, 5 = mid-range 1 = expensive	2	9	6	3
Power Usage	TCS - 06	10 = low, 5 = medium 1 = high	2	5	6	9
Mass	TCS - 08	10 = low mass, 5 = medium 1 = high mass	3	10	6	3
Size	TCS - 08	10 = very small, 5 = medium 1 = very large	3	10	7	3
Thermal Controllability	TCS - 01	10 = precise, 5 = medium 1 = imprecise	5	8	9	2
Reliability + TRL	TCS - 01, TCS - 04	10 = high, 5 = medium 1 = low	3	9	8	6
TOTALS:				155	132	70

This trade study evaluates potential options for generating heat for the system. Three options were evaluated: patch heaters, cartridge heaters, and a deployable surface to increase solar flux on the system. The first two options are two different forms of electrical resistance heaters and the deployable surface would be a mechanical system which can increase the size of the top surface of the system, allowing for an increased level of solar flux. The criteria used to evaluate these options are: cost, power usage, mass, size, thermal controllability, and reliability + TRL.

Each criteria was given a weight on a scale of one to five, with one being least important and five being most important. Cost and power usage were given a weight of 2 because although it is important to minimize both, a system with higher cost or power usage might be worth accepting if it scores well in other categories. Mass and size are similar, but they were given a weight of 3 because they are more tightly constrained, especially after the newest mass and size constraints. Reliability and TRL is important as well because a failure in heat generation would be extremely problematic, but the requirement TCS-04 ensures redundancy in heat generation, so a weight of 3 was selected for this criteria. Finally, thermal controllability is vital because it ensures that the system will generate the appropriate amount of heat: enough heat to reach the operating temperature ranges of all components without overheating them.

Patch heaters are lightweight, thin heaters which generate heat through electrical resistance. Their small size and mass earned them a score of 10 for both of those criteria. They scored a 9 for cost because they are produced commonly and do not require any difficult manufacturing to produce. For power usage, they scored a 5 because the heat they generate is roughly equal to the power they use. They scored an 8 for thermal controllability because they can be run at different power levels to generate different quantities of heat, but their flat shape means that they are less efficient at heating the inside of components. Finally, they scored a 9 for reliability + TRL because they have been used frequently in previous space applications, and their use for this mission would be similar to their use for previous missions.

Cartridge heaters are similar to patch heaters, generating heat from electrical resistance, but they are cylindrical and generate the heat inside a metal sheath. They are much more costly to manufacture than patch heaters, and their shape makes them heavier and larger, so they scored a 6 for both cost and mass and a 7 for size. However, they can provide heat more directly and precisely than a patch heater, so they scored slightly better in power usage and thermal controllability, at a 6 and a 9, respectively. Finally, these have also been used in various spacecraft as well, but not as frequently as patch heaters, so they scored an 8 for reliability.

Finally, the deployable surface did not score well. With its high manufacturing cost, mass, and size, it scored only a 3 for each of those criteria. It scored a 9 for power usage because it would only need power to deploy and retract the surface; however, the heat generation would be very inconsistent because it relies on solar flux, which will not be present at night or in caves, so it scored a 2 for thermal controllability. It is a fairly simple mechanical system and would likely be fairly reliable to deploy and retract as intended, but is not commonly used in other applications, so it scored a 6 for reliability + TRL.

This trade study revealed that patch heaters are the best method of heat generation for this project, so the TCS will be designed to use them.

Heat Distribution Method						
Criteria	Explanation	Grade	Weight: Scale {1-5}	Fluid Loops	Thermal Straps	None
Cost	TCS - 02	10 = cheap, 5 = mid-range 1 = expensive	2	2	8	10
Power Usage	TCS - 06	10 = high, 5 = medium 1 = low	2	3	10	10
Mass	TCS - 08	10 = low mass, 5 = medium 1 = high mass	3	3	8	10
Size	TCS - 08	10 = very small, 5 = medium 1 = very large	3	4	6	10
Distribution Consistency	TCS - 01	10 = very consistent throughout system 5 = medium 1 = large temperature differences likely	5	10	9	1
TOTALS:				81	123	105

In deciding the heat distribution method, five criteria were used: cost, power usage, mass, size, and distribution consistency. The weights for these, respectively, are 2, 2, 3, 3, and 5. Cost, power usage, mass, and size all need to be minimized, so each is important, but with the new restrictions on mass and size they were assigned a slightly higher weight than cost and power usage. Distribution consistency is weighted at a 5 because it is the most important factor, and it determines how well the option distributes heat throughout the system to ensure that the system has a consistent internal temperature.

Fluid loops are the first option considered for this subassembly, and they consist of pumping fluid from a heat source to other parts of the system. Fluid loops scored poorly for each of the constrained categories, with a 2 for cost, a 3 for power usage and mass, and a 4 for size. These low scores are due to their large size and required power draw compared to other systems. For distribution consistency, they scored a 10 because of their proven ability to transfer heat from one part of a system to another.

Thermal straps are a passive heating element which are designed to conduct heat between any elements they connect. They scored an 8 for cost and mass because they are fairly easy to manufacture and not overly heavy, and because they are a passive element, they don't use any power, so they scored a 10 for power usage. For size, they scored a 6 because they do take up some space within the system, but are not especially bulky. Finally, they scored a 9 for distribution consistency because they are also a proven method of heat transfer, though less control over them is available compared to fluid loops.

The final option considered was having no dedicated heat distribution system, instead relying on conduction through the chassis and convection through the surrounding air. This scored a 10 for cost, power usage, mass, and size because it would take nothing in all of those categories. However, it scored a 1 for distribution consistency because there is no method in the system to ensure that components are heated equally, so large temperature gradients within the system would be very likely should this be the option used.

Despite the benefits of the other two options, thermal straps scored the highest overall thanks to consistently high scores for all categories, so they will be used in our system.

Interface(s) with the Environment						
Criteria	Explanation	Grade	Weight: Scale {1-5}	White radiator on top face, MLI covering all other faces	MLI on all faces	None, leave system fully exposed to environment
Reduction in Power Usage	TCS - 06	10 = significant reduction 5 = medium 1 = no reduction	3	8	10	1
Reduction in Overheating Risk	TCS - 01, minimizing risk	10 = significant reduction 3 = no reduction 1 = increases risk	4	8	1	3
Cost	TCS - 02	10 = cheap, 5 = mid-range 1 = expensive	1	4	5	10
Mass	TCS - 08	10 = low mass, 5 = medium 1 = high mass	2	7	7	10
Size	TCS - 08	10 = very small, 5 = medium 1 = very large	2	7	7	10
TOTALS:				88	67	65

The final decision needed to be made is how our system will interface with the surrounding environment. To make this decision, five criteria were used: reduction in power usage, reduction in overheating risk, cost, mass, and size. These criteria were weighted on a one to five scale, and the following weights were given: 3 for reduction in power usage, 4 for reduction in overheating risk, 1 for cost, and 2 for mass and size. The first two were the highest weights because they encompass the actual goal of this subassembly, but a high risk of overheating is more detrimental to the system than a high power use.

The first option is painting the top face white to act as a radiator while covering all the other faces in multi-layer insulation (MLI). This scored an 8 in both reduction of power usage and reduction of overheating risk because it minimizes the overall heat flux throughout the system while leaving one face open to the environment, radiating heat outwards. It scored just a 4 for cost due to needing MLI and likely needing to restructure the top face so it could be covered in white paint, but it scored a 7 for both mass and size due to the fairly low mass and size of MLI and paint.

The second option is fully enclosing the system in MLI. This scored a 10 for power reduction because it drastically minimizes heat flux through the system, but it scored only a 1 for reduction in heating risk because there is no method for fast cooling, so it would actually increase that risk. It scored a 5 for cost due to the cost of MLI, but a 7 for mass and size for the same reason as the first option.

The final option is to leave the system fully exposed to the environment without any dedicated thermal interface. It scored a 1 for reduction in power usage and 3 for reduction in overheating risk because it does nothing for either of these, though it doesn't increase the risk either. Despite this, it scores a 10 for all three of cost, mass, and size.

The trade study shows that the best option for the thermal interface with the environment is putting a white radiator on the top face and MLI on all other surfaces.

One paragraph for recovery (software) and one for redundancy (hardware)

Recovery

The TCS will rely on software to tell the heaters when to turn on and off based on data collected by temperature sensors. Based on the operational parameters for the scientific instruments, we will have an internal temperature goal of 20 degrees, so if any of the temperature sensors read under that goal, the software should turn on the heater corresponding to that sensor, and if the temperature reaches that mark or is above it, the heater should turn off.

Redundancy

Throughout the TCS, there will be multiple redundant systems which will help in case of failures or malfunctions. Each patch heater will have a backup, and "shortcut routes" for heat distribution will be added with the thermal straps, so no component or subsystem is completely reliant on one thermal strap. The extra patch heaters are true backups, and will only be put in use if the

originals break, but the extra thermal straps will help distribute temperature more quickly even when all thermal straps are operational. For the environment interface, the only redundancy will be at connection points, ensuring that one faulty connection will not compromise the system.

At least two risks for inclusion in the risk matrix

1. Overheating
 - Our system has no active cooling, so the only way to cool is to use less power and radiate heat into the environment
2. Failed heaters
 - Unlikely due to the presence of redundant heaters, but would be very problematic if it happens
3. Incorrect sensors (part CDH)
 - If the sensors tell the TCS the incorrect temperature, we could either overheat or not heat the system enough

Sources:

[Mars Climate Database](#)